# ALIYAN FAISAL

AI Engineer & Aspiring Researcher

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cuelara.com · invoiceinspect.app

## RESEARCH PROFILE

AI Engineer and aspiring researcher with a BSc in Information Technology (CGPA 3.80/4.00), specializing in applied AI systems, LLM orchestration, and Retrieval-Augmented Generation (RAG) architectures. Experienced in building scalable data pipelines, prompt engineering toolkits, and intelligent decision-support platforms. Strong foundation in computer science fundamentals, backend system design, and API-driven automation. Seeking admission into a research-oriented Master's program in Artificial Intelligence or Machine Learning to move from building LLM-based systems to researching their reliability, evaluation and efficiency.

## RESEARCH INTERESTS

- **LLM Reliability, Uncertainty & Calibration:** Measuring correctness, and when to say "could not verify".
- **Evaluation Methodology:** Detecting and reporting score inflation from tuning on test sets.
- **Efficient & Small Language Models:** Best quality from small or quantized models on commodity hardware.
- **Retrieval-Augmented Generation:** How parsing, chunking and retrieval affect answer faithfulness.
- **Prompt Optimization, Token Efficiency & Tokenization Across Languages:** How much prompt can go before quality drops; why token cost differs by language.
- **Multimodal Document Understanding & Data-Centric AI:** Combining text and vision readings.
- **Learned vs. Rule-Based Information Extraction:** When learned models beat rules on unseen layouts.
- **Machine Learning for Prompt Evaluation & Optimization:** What can be learned, and how to validate scores.
- **AI for Climate & Disaster Resilience in Mountain Regions:** Low-cost early warning where data is sparse.

## EDUCATION

**Karakoram International University** Gilgit, Pakistan
*Bachelor of Science in Information Technology* **CGPA: 3.80 / 4.00** (2019 – 2023; result declared May 2024)

- **Class Rank:** Ranked 1st in class in the BSIT program.
- **Scholarship:** HEC Ehsaas Scholarship (fully funded for the entire degree).
- **Core Coursework:** Data Structures & Algorithms, Discrete Structures, Linear Algebra, Probability & Statistics, Database Systems, Operating Systems, Computer Network, Information Security, Software Engineering.

## TECHNICAL SKILLS

- **AI Engineering & RAG:** LLM APIs (OpenAI, Gemini, Anthropic, Groq, DeepSeek), Local LLMs, Tool and Function Calling, Retrieval-Augmented Generation (RAG), Vector Embeddings, Prompt Optimization, Context Window Management, Token Reduction.
- **Languages & Backend:** Python, PHP (Laravel), Node.js, Flask, TypeScript, JavaScript, RESTful APIs, SQL.
- **Data & Infrastructure:** Pandas, NumPy, PostgreSQL, MySQL, MongoDB, Vector Databases, Docker, Git, Linux Administration.

## FEATURED AI & SOFTWARE PROJECTS

**InvoiceInspect – Multimodal Invoice Verification & Error Detection System** invoiceinspect.app
*Creator & Lead Systems Engineer* **2025 – Present**

- Built a system that finds arithmetic and consistency errors in PDF invoices (totals, tax, duplicate charges, missing fields) and reports each with evidence: expected value, printed value and page.
- Designed dual-path extraction (text/OCR with Tesseract plus a vision model), cross-checked field by field; disagreements are marked "could not verify" instead of guessed.
- Kept verification deterministic: a rules engine decides and the LLM only explains findings, with provider fallback.
- *In development:* RAG chat on the results page to ask questions about a verified invoice and its findings.
- Built an evaluation harness and published a research report and eight test reports. On 30 unseen invoice layouts, the rules extractor scored 70.3% field accuracy and the vision extractor **97.9%**. After tuning, rules reached 100% on 10 held-out invoices (small sample); I now report tuned and untuned scores separately after tuning inflated an earlier 99.9%.

**Cuelara – LLM Prompt Optimization & Context Engineering Suite** cuelara.com
*Creator & Lead Systems Engineer* **2025 – Present**

- Built a toolkit with a **Prompt Optimizer**, **Token Reducer**, **Context Extractor** and **Prompt Debugger** to cut context-window overhead and improve response consistency.

- Engineered token-reduction algorithms, formatting rules and context extraction that separates system instructions from input noise.
- Built the **Document to Prompt** tool using Retrieval-Augmented Generation (RAG): relevant passages are retrieved from an uploaded document and turned into a focused prompt.
- Shipped as a web app, browser extension, public API and MCP server, with plan-based rate limiting.

**ManaJet – AI-Assisted Project & Task Management for Small Software Teams** **Final Year Project**
*Lead System Developer (team of two)* **2023**

- Built an AI task planner (OpenAI APIs) that generates task breakdowns, estimates and dependencies, with output constrained by JSON schema and validated before storage.
- Built the REST API and MySQL schema for project state and task dependencies, together with one teammate.

## OPEN SOURCE

**Notemind – Claude Code Plugin for Notes and Reminders** github.com/aliyanfaisal/notemind
*Author* **MIT License**

- Free plugin to add notes in plain words from a Claude Code chat (local SQLite), with a session-start hook (about 85 ms) reporting overdue notes.
- Least-privilege design (only the user can add or close notes, never the model); no network calls.

## EXPERIENCE

**UrbanSofts – Remote Software Development Team** Remote
*Founder & CEO* **2023 – Present**

- Founded and lead a remote team of 4–10 developers: I take on client projects and bring in the right members for each, and we build them together. The team has served 50+ clients.

**Planning & Development Department, Government of Gilgit-Baltistan** Gilgit, Pakistan
*Backend Developer & Data Analyst (Voluntary Work)* **Feb 2025 – Jul 2025**

- Built centralized data analytics dashboards processing public-sector records (XLS, CSV, MySQL) using Python (Pandas) and Laravel backend.
- Digitized administrative paper workflows into structured reporting frameworks to aid decision-support systems for regional planning.

**Greymatter Ventures Pvt** Gilgit, Pakistan
*Full Stack Web Developer (Team Lead), Part-time* **Jan 2021 – Dec 2023**

- Designed modular backend architectures, relational schemas, and server-side logic to optimize application throughput and API reliability.
- Led the web development team, mentoring junior developers and remote interns on code optimization, system design, and deployment workflows.

**Freelance (Upwork & Fiverr)** Remote
*Applied AI & Backend Engineer* **2020 – Present (5+ years)**

- Engineered production-grade applications integrating LLM APIs (OpenAI, Claude, Gemini) for automated document summarization, data extraction, and dynamic e-commerce processing.

## LANGUAGES

**Urdu:** Native · **English:** Professional working proficiency

## REFERENCES

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